

Qoralis Interior and Surface Explorer (QISE) - CADRe Part A

Office of Mission Cost Analysis (OMCA)

Cost Analysis Data Requirements (CADRe)

PART A

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1 INTRODUCTION

1.1 PURPOSE OF THIS DOCUMENT

CADRe Part A provides a concise, standardized synopsis of the mission or project to support costing and analysis activities. Its primary purpose is to consolidate technical, programmatic, operational, and schedule information into a single normalized reference.

Specifically, CADRe Part A:

- Summarizes key mission characteristics in a consistent and structured format
- Aggregates information from the QISE mission concept, preliminary requirements, architecture, schedule, and procurement assumptions
- Reduces the need for analysts to repeatedly consult disparate source material when constructing cost and risk estimates
- Ensures a common baseline description is used across cost, schedule, technical, and risk analyses
- Improves traceability between quantitative mission drivers and downstream cost-estimating assumptions

In practical terms, CADRe Part A functions as a time-saving reference document that enables analysts to understand the mission without re-interpreting the architecture or reconciling multiple preliminary sources.

1.2 PURPOSE AND SCOPE OF MISSION

Qoralis Interior and Surface Explorer (QISE) is a fictitious deep-space planetary science mission intended to determine the internal structure, shallow subsurface properties, and surface thermal behaviour of Qoralis, a synthesized 410 km-diameter icy dwarf body in the outer solar system. The mission is sponsored, owned, and operated by the National Space Exploration Office (NSEO), with scientific direction from the fictitious Planetary Geophysics Directorate (PGD).

QISE responds to a defined science need: determine whether Qoralis contains a mechanically distinct interior layer or long-lived subsurface liquid reservoir, quantify the thickness and dielectric properties of the upper 5 km of crust, and establish the relationship between surface fractures and internal heat transport.

The mission uses two coordinated spacecraft with deliberately different functions: an Orbiter (QO-1) performs global geophysics and remote sensing from a polar orbit, while a Surface Geophysics Lander (QS-1) performs local seismic, thermal, and near-surface measurements. The two-spacecraft architecture separates global orbital measurements from long-duration surface measurements while retaining a common mission operations and data system.

The project is entirely synthesized for cost-analysis purposes. No existing spacecraft, launch vehicle, instrument, company, government agency, or previously flown mission is assumed as the technical baseline. Generic engineering practices are used only to make the architecture internally consistent; all organization names, spacecraft names, target-body properties, performance allocations, and program dates in this CADRe are fictional.

2 GENERAL DESCRIPTIVE INFORMATION

2.1 MISSION OVERVIEW

The Qoralis Interior and Surface Explorer (QISE) is a proposed two-spacecraft deep-space mission designed to characterize the interior and surface of the fictional icy body Qoralis.

Qoralis is defined for the QISE reference case as a 410 km mean-diameter body with a bulk density of 1.25 g/cm^3 , a nominal rotation period of 19.6 hours, and a heliocentric orbit with a semi-major axis of 3.35 AU and eccentricity of 0.07. The reference body model is synthetic and exists only to provide stable mission-analysis inputs.

The QISE spacecraft will travel together during cruise and separate approximately 45 days before target arrival. QO-1 will enter a $100 \times 1,500$ km polar elliptical science orbit around Qoralis; QS-1 will perform a controlled descent to a surface landing site between 65°S and 70°S latitude selected for high fracture density and communications visibility.

The mission's broad science objectives include:

- Global gravity-field mapping and determination of mass distribution;
- Measurement of surface topography and long-wavelength shape;
- Detection and characterization of subsurface dielectric interfaces to a depth of at least 5 km;
- Measurement of local seismic activity and elastic-wave propagation;
- Mapping of thermal anomalies with a target sensitivity of 0.8 K at the instrument level; and
- Correlation of orbital observations with long-duration surface measurements.

The system consists of one orbital geophysics spacecraft, one surface geophysics lander, a combined launch service, government and contractor-provided ground infrastructure, mission-control and planning systems, deep-space communications, science-data processing and archive capability, and associated project-management, assurance, and sustainment activities.

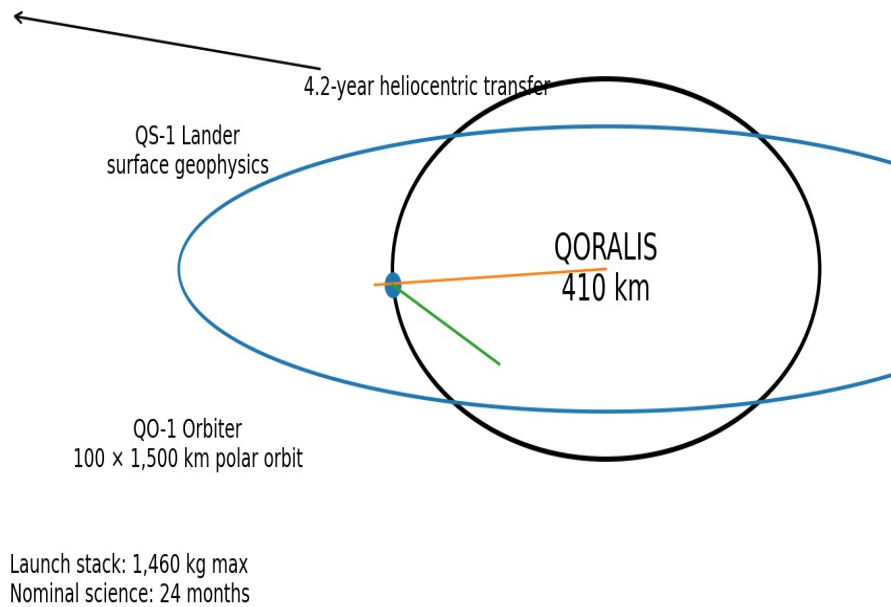
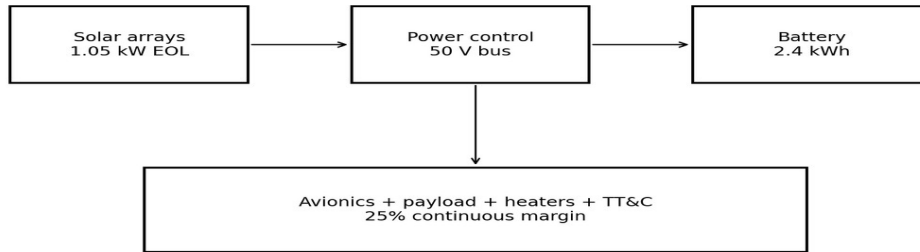


FIGURE 1 – QISE MISSION CONCEPT



Avionics + payload + heaters + TT&C
25% continuous margin



2.2 CONTEXT

The QISE concept is not derived from a named real mission. Its programmatic context is a notional national planetary-science initiative requiring a flight system capable of operating beyond 3 AU with a surface asset and an independently navigable orbiter.

The target body is intentionally defined as a fictional object so that the cost baseline is not constrained by an existing mission design, real spacecraft heritage, or commercial provider. The synthetic environment nevertheless imposes realistic engineering drivers: approximately 4.2 years of cruise, low solar flux, a 7.5-year total program duration from launch through nominal disposal, deep-space communications, radiation exposure, autonomous fault management, and a precision landing sequence.

- QISE retains no procurement baseline from real industry responses. The reference organizations used in this CADRe are fictitious: Heliox Orbital Systems for the orbiter bus, Cinderline Surface Systems for the lander, Blue Meridian Instruments for the science payloads, Quarry Mission Data for the ground system, and Vantor Launch Services for the launch service.
- The QISE CADRe baseline in this document is the controlling Part A reference for the cost estimate. Any change to the target-body model, spacecraft architecture, transfer trajectory, payload allocation, schedule, or organizational baseline shall be reconciled across the later CADRe products.
- Reference Industry Responses – Not QISE Procurement Baseline
- Reference supplier / configuration observations are intentionally generic and fictional. They are retained only as placeholders for cost-estimating structure and do not represent actual supplier quotations.
- mission assurance and long-duration spacecraft operations are not inherited from any real mission; all QISE architecture and program assumptions are synthetic.
- No external mission is treated as a technical or cost analogue. Quantitative values are generated from the QISE reference-body model and internal mission trades.

Reference Industry Responses – Not QISE Procurement Baseline

Prime	Architecture	Bus / Element	Payload / Role
Heliox Orbital Systems	2 spacecraft	QO-1 orbital bus	Orbiter avionics integration
Cinderline Surface Systems	1 lander	QS-1 surface bus	Landing and surface systems
Blue Meridian Instruments	Orbiter + lander payloads	Instrument packages	Radar, thermal, seismic
Quarry Mission Data	Ground segment	Mission data infrastructure	Processing / archive / comms integration
Vantor Launch Services	Launch service	Combined launch stack	Launch and separation

2.3 STRATEGIC OBJECTIVES

Consistent with the NSEO Exploration Strategy, QISE is intended to obtain quantitative geophysical measurements of Qoralis that cannot be obtained from Earth-based observations alone, establish a traceable relationship between orbital and surface datasets, and demonstrate autonomous deep-space operations for a multi-spacecraft planetary mission.

TABLE 1 – STRATEGIC ALIGNMENT

	Strategic Goals	Exploration Strategy
1	MEASURE	QISE will establish quantitative measurements of the interior, subsurface, and surface thermal state of the fictional body Qoralis, linking orbital and surface observations.

2.4 SELECTED MISSION REQUIREMENTS

The following table consolidates the quantitative mission parameters that are significant drivers of QISE cost, schedule, risk, and technical complexity. Values are a preliminary estimating baseline; unresolved items are intended to close through the design reviews identified in Section 6.

TABLE 2 – SELECTED MISSION REQUIREMENTS

Area	Key requirement
Target body	Synthetic 410 km-diameter icy body; reference density 1.25 g/cm ³ and rotation period 19.6 h.
Cruise	Launch-to-arrival planning duration of approximately 4.2 years; trajectory correction campaign allocated 360 m/s.
Orbiter orbit	QO-1 polar elliptical science orbit of 100 × 1,500 km around Qoralis; final orbit insertion profile TBD by navigation analysis.
Lander landing	QS-1 shall land within a designated 20 km × 20 km science ellipse with terminal horizontal velocity ≤1.5 m/s.
Mission duration	Nominal science operations of 24 months after commissioning; total launch-to-disposal mission duration approximately 7.5 years.
Combined launch mass	≤1,460 kg including flight hardware, propellant, adapter allocation and growth margin.
QO-1 mass	≤720 kg at launch; preliminary target 650 kg including propellant.
QS-1 mass	≤430 kg at separation; preliminary target 400 kg including landing propellant.
Payload mass	QO-1 payload ≤95 kg; QS-1 payload ≤42 kg.
QO-1 power	≥1.05 kW end-of-life solar generation at 3.35 AU; ≥25% continuous power margin.
QS-1 power	≥180 W end-of-life surface generation; ≥30% survival-mode margin.
Radar sounding	8–20 MHz operating band; minimum investigation depth 5 km; nominal vertical resolution ≤50 m.
Thermal mapping	7–14 μm band; instrument NETD ≤0.15 K at 250 K; ≤250 m ground sampling at reference orbit.
Seismic measurement	Broadband surface seismometer with timing accuracy ≤1 ms and continuous nominal sampling.
Altimetry	QO-1 laser altimeter range precision ≤0.5 m for 1 km–1,500 km slant range under reference conditions.
Pointing	QO-1 absolute pointing error ≤0.015° during radar mapping; ≤0.03° during thermal mapping.
Orbit knowledge	QO-1 3-sigma position knowledge target ≤250 m during nominal science operations.

Area	Key requirement
Onboard storage	QO-1 ≥ 512 GB usable; QS-1 ≥ 128 GB usable non-volatile storage.
Science data	Planning case of 8 GB/day QO-1 and 1.2 GB/day QS-1 before compression.
Downlink	Ka-band science downlink target ≥ 2 Mbps peak with ≥ 3 dB end-to-end link margin.
Priority latency	Priority science products available within 72 hours of receipt by the ground segment.
Autonomy	QO-1 ≥ 24 h autonomous operation; QS-1 ≥ 72 h autonomous surface operation.
Ground availability	Deep-space ground system $\geq 98\%$ availability during scheduled communications periods.
Navigation	Radiometric plus optical navigation; final arrival solution shall close before first science orbit.
Reliability	Preliminary 7.5-year mission-success probability target ≥ 0.75 for cost/risk analysis.
Mass margin	$\geq 15\%$ nominal mass margin at system preliminary design baseline.
Power margin	$\geq 25\%$ QO-1 nominal continuous margin and $\geq 30\%$ QS-1 survival-mode margin.
Technology readiness	Major bus components TRL ≥ 8 before qualification; critical payload technologies TRL ≥ 5 at B2 start and ≥ 7 before spacecraft-level qualification.
Landing readiness	Landing system shall complete a representative end-to-end descent rehearsal before launch.
End of life	QO-1 shall be placed in a non-interfering disposal orbit or controlled trajectory; QS-1 remains on the surface after completion.

2.5 HIGH-LEVEL MISSION ARCHITECTURE

Owned and operated by NSEO, in collaboration with the Planetary Geophysics Directorate, the mission consists of:

- A Program Segment: NSEO project management, systems engineering, science management, mission assurance, configuration control, mission planning, and government technical authority.

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- A Space Segment: one orbital spacecraft (QO-1) and one surface geophysics lander (QS-1), launched as a combined stack and separated before target arrival.
 - A Launch Segment: a commercial launch service procured for the combined QISE stack; Vantor Launch Services is the current fictional launch-service organization in the baseline.
 - A Ground Segment: contractor-provided deep-space communications, navigation support, mission planning, command, science-data processing, calibration, archive, and secure distribution. Quarry Mission Data provides and integrates these elements.
 - An Operations Segment: NSEO mission operators supported by contractor flight-dynamics and engineering teams. Nominal operations require 16-hour daily staffed coverage with on-call anomaly response outside staffed periods.
 - A Science Segment: the PGD science team and fictional university laboratories supporting instrument calibration, geophysical inversion, landing-site analysis, data validation, and scientific exploitation.

3 SPACE SEGMENT DESCRIPTION

The space segment consists of 2 spacecraft: one orbiter and one surface geophysics lander, each with a dedicated bus and payload complement.

3.1 SPACECRAFT BUS

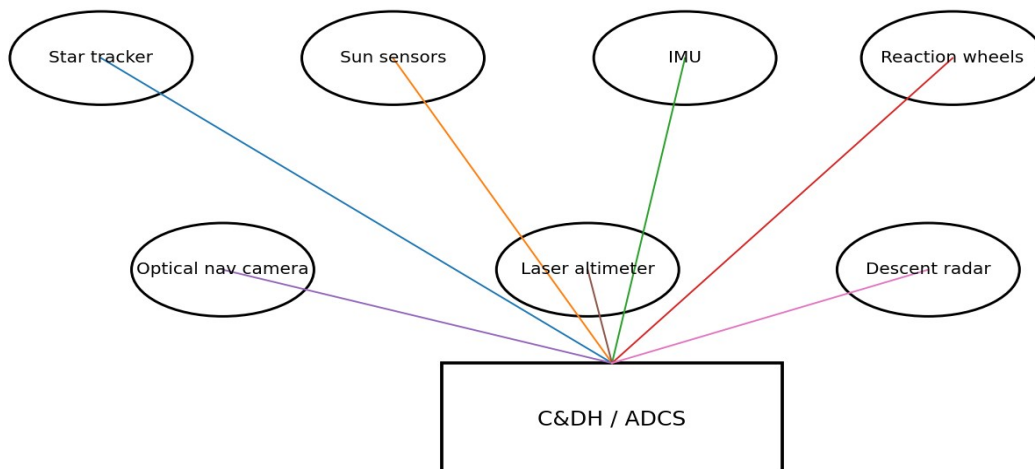
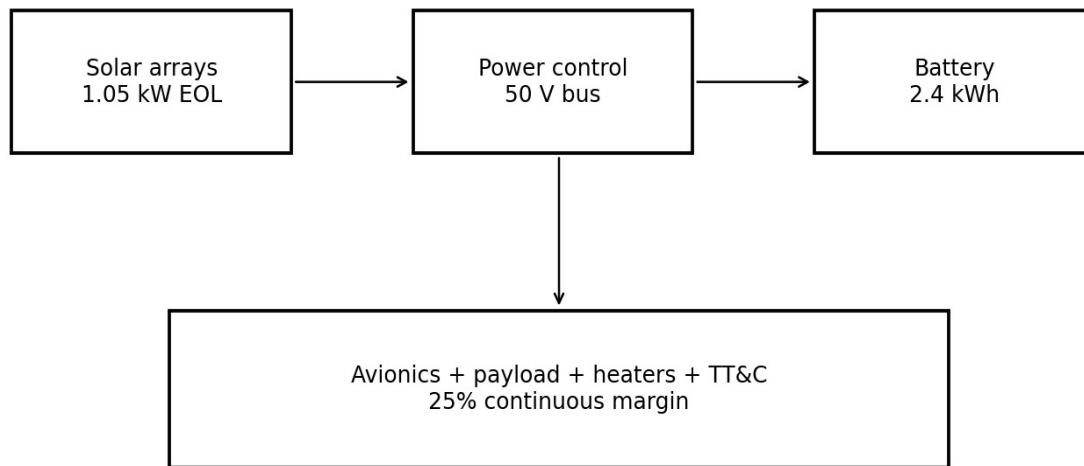
The QISE spacecraft bus architecture provides the structural, electrical, thermal, computational, communications, attitude-control, navigation, and propulsion functions required for a 3.35 AU-class heliocentric cruise and subsequent operations around Qoralis. The orbiter and lander use different mechanical and thermal configurations but share a common command/data protocol, time system, fault-management philosophy, and ground interface.

The orbiter and lander are intentionally not identical. QO-1 is optimized for low-propellant orbital operations, precision pointing, high-rate science downlink, and global mapping. QS-1 is optimized for entry/descent/landing survivability, surface stability, thermal endurance, and continuous low-rate geophysical measurements. A single Engineering Qualification Unit (EQU) is used for the highest-risk common avionics and communications interfaces; each flight vehicle then undergoes independent acceptance.

Key functions of the spacecraft buses include:

- Structural subsystem
- Electrical power generation and distribution
- Command and Data Handling (C&DH)
- Thermal subsystem
- Attitude Determination and Control System (ADCS)
- Propulsion and navigation subsystem
- Telemetry, Tracking and Command (TT&C)

All component examples in this section are fictional notional hardware used only for first-order costing. No real spacecraft, instrument, or commercial product is treated as QISE flight heritage.



3.1.1 Structural Subsystem

The QO-1 primary structure uses a machined aluminum-lithium frame with carbon-fibre instrument decks and local titanium fittings. QS-1 uses a low-centre-of-gravity landing frame with crushable load paths and four deployable landing legs. Structural sizing shall demonstrate a minimum 1.25 ultimate factor on limit loads and a first-mode frequency above 28 Hz for the integrated launch configuration.

The preliminary structure allocation is 92 kg for QO-1 and 68 kg for QS-1, including primary structure, secondary panels, fasteners, launch restraints, landing hardware, and 20% design mass margin at the applicable baseline.

Parameter	Preliminary QISE allocation / target
QO-1 launch mass	≤720 kg; target 650 kg
QS-1 separation mass	≤430 kg; target 400 kg
Combined flight-system mass	≤1,150 kg before launch adapter; target ~1,050 kg
Launch-stack allocation	≤1,460 kg including adapter and growth margin
QO-1 payload allocation	≤95 kg
QS-1 payload allocation	≤42 kg
QO-1 usable science storage	≥512 GB
QS-1 usable science storage	≥128 GB
QO-1 pointing	≤0.015° radar / ≤0.03° thermal
QS-1 landing ellipse	20 km × 20 km
QO-1 total ΔV	780 m/s preliminary budget
QS-1 total ΔV	420 m/s preliminary budget
QO-1 EOL power	≥1.05 kW at 3.35 AU
QS-1 EOL power	≥180 W surface generation
Qoralis reference orbit	100 × 1,500 km polar science orbit

3.1.2 The electrical-power systems shall close the end-of-life power budget with at least 25% continuous power margin for QO-1 cruise/science modes and at least 30% margin for QS-1 surface survival mode. The design shall account for solar-distance degradation, battery ageing, heater duty cycle, communications loads, instrument duty cycle, and peak landing loads.

Solar Array Panels

- QO-1 will use two body-deployed solar wings sized for operation at 3.35 AU. Preliminary array area is 38 m² with an end-of-life generation target of 1.05 kW at the reference heliocentric distance. QS-1 will use a 5.5 m² fixed deployable array with a 180 W end-of-life target during the surface science phase.
- End-of-life power generation shall be sufficient to meet the verified mission power budget with the specified design margin. Solar-cell degradation is preliminarily allocated at 3.0% per year for QO-1 and 2.0% per year for QS-1 after landing.

- The power system shall be sized for deep-space communications, thermal control, instrument operation, battery recharge, avionics, propulsion valves, and landing-event peak loads.

Battery

- QO-1 uses a 2.4 kWh lithium-ion battery with a preliminary 80% usable-depth-of-discharge limit. QS-1 uses a 0.9 kWh battery sized for 18 hours of surface survival without solar input. Final capacities are TBD pending thermal and power-budget closure.
- Power Control Unit

The power-control and distribution architecture shall provide regulated power, protection, load shedding, telemetry, and autonomous load management. Peak bus voltage is nominally 50 VDC on QO-1 and 28 VDC on QS-1.

- The power-control and distribution architecture shall provide regulated power, protection, telemetry, load shedding, and autonomous load management for all spacecraft and payload functions.

QO-1 Solar Wings 38 m ² reference area	QO-1 Power Control Unit 50 V regulated bus	Battery Pack 2.4 kWh usable allocation
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FIGURE 2 – NOTIONAL SPACECRAFT POWER-SYSTEM EXAMPLES

3.1.3 The Command and Data Handling (C&DH) subsystem provides central computing, timekeeping, fault management, data storage, instrument control, and communications-interface functions for each spacecraft.

The C&DH subsystem will:

Receive and execute time-tagged commands;

Collect and compress engineering telemetry;

- Control payload interfaces and instrument modes;
- Manage onboard science-data storage;
- Execute fault-detection, isolation, and recovery;
- Maintain spacecraft configuration and software versions;
- Support autonomous operation during communication outages; and
- Maintain synchronized mission time across QO-1 and QS-1.
- QO-1 shall provide ≥ 512 GB usable non-volatile science storage; QS-1 shall provide ≥ 128 GB, for 640 GB combined.
- Fault Management

-
- The C&DH architecture will provide watchdogs, health monitoring, fault isolation, safe-mode entry, redundant-unit selection, autonomous load shedding, command validation, and recovery after loss of ground contact.
 - Data Management
 - The C&DH subsystem will buffer radar, thermal, altimetry, seismic, and surface-monitoring data before transmission. Design inputs include generation rate, compression, storage, communication windows, priority data, and round-trip latency.

Because the two spacecraft will operate as a coordinated constellation, configuration and software management shall ensure that both spacecraft remain compatible with the common mission architecture, software baseline, time reference, operational procedures, and mission-planning interfaces.

3.1.4 Thermal Control Subsystem

The Thermal Control Subsystem will maintain spacecraft and payload components within qualified temperature limits throughout cruise, Qoralis orbit operations, and the surface phase. Thermal design is dominated by low solar flux, long eclipse periods around the target, internal heater loads, and the need to keep the radar electronics and surface seismometer within stable operating ranges.

Thermal analysis will therefore consider:

Solar radiation

- Planetary/body infrared radiation
- Albedo
- Eclipse duration
- Orbital position
- Spacecraft attitude
- Instrument dissipation
- Communications equipment
- Battery temperature
- Propulsion components
- Lander surface-contact thermal conductance
- The thermal architecture will use passive thermal-control techniques wherever practical. Potential components include multi-layer insulation, optical coatings, radiators, thermal straps, heat pipes, heaters, thermostats, and temperature sensors.

Thermal Analysis

The spacecraft will undergo hot-case and cold-case thermal analyses for cruise, target arrival, orbital operations, and the surface phase. The analysis will establish component temperature ranges, heater requirements, radiator sizing, thermal gradients, instrument stability, battery limits, and landing-leg thermal constraints.

Cost Drivers

- Thermal-system cost will be affected by payload heat dissipation, radiator area, heater power, thermal stability requirements, low-flux environmental analysis, instrument duty cycle, and environmental testing duration.

3.1.5 Attitude Determination & Control

QO-1 coarse attitude determination will use sun sensors and rate gyros; fine attitude determination will use two star trackers. Three-axis control will use four reaction wheels with magnetic torquers for momentum management during cruise and target operations.

- QS-1 uses redundant inertial measurement units, sun sensors, and a radar altimeter during descent. After landing, attitude knowledge is established from inertial sensors and horizon/solar observations; active attitude control is limited to controlled instrument pointing and lander health management.
- QO-1 absolute pointing accuracy target is $\leq 0.015^\circ$ during radar mapping and $\leq 0.03^\circ$ during thermal imaging. QS-1 instrument deck pointing knowledge shall be $\leq 1.0^\circ$ after landing.
- Orbit determination for QO-1 will use optical navigation observations plus radiometric tracking. A preliminary 3-sigma position-knowledge target of ≤ 250 m at Qoralis is retained for science operations.
- Orbit Determination
- QO-1 orbit determination shall combine two-way range, Doppler, onboard time, and optical landmark measurements. The final navigation filter shall close before the first science orbit.
- Navigation sensors

QO-1 will carry a star tracker pair, sun sensors, inertial measurement unit, and optical navigation camera. QS-1 will use an inertial measurement unit, descent radar, and redundant altimeter channels during landing.

- Control actuators
- QO-1 will use four reaction wheels and redundant hydrazine-equivalent fictional monopropellant thrusters; QS-1 will use twelve landing/descent thrusters arranged in four functional clusters.

Star tracker	Sun sensor
Inertial measurement unit	Reaction wheel

Optical navigation camera	Magnetorquer
Laser altimeter	Descent radar / altimeter

3.1.6 Propulsion subsystem

QISE propulsion is sized for a 4.2-year cruise correction campaign, QO-1 orbit insertion and shaping, and QS-1 descent and landing. The preliminary propellant budget includes 20% contingency.

Parameter	QO-1 Value	QS-1 Value
Propellant Type	Fictional high-density monopropellant	Fictional storable monopropellant
Vacuum Specific Impulse (Isp)	228 s	225 s
Nominal Thrust	22 N	18 N
Tank capacity	185 kg	96 kg
kg	20%	

Propulsion System Characteristics

Item	Value
Cruise correction	360 m/s
Qoralis orbit insertion / shaping	290 m/s
Station keeping	70 m/s
Disposal contingency	60 m/s
Required QO-1 ΔV	780 m/s
QS-1 descent / landing reserve	420 m/s

Notes

- QO-1 total mission delta-V is 780 m/s, including 360 m/s cruise correction, 290 m/s target orbit insertion and shaping, 70 m/s station keeping, and 60 m/s disposal

contingency. QS-1 requires 420 m/s for deorbit, descent, terminal hazard avoidance, and landing reserve.

- QO-1 trajectory corrections are designed from the synthetic Qoralis reference orbit; no real-body station-keeping or launch-window heritage is assumed.

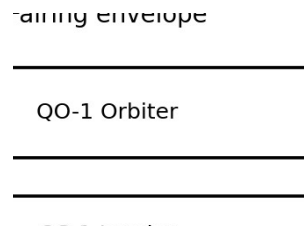
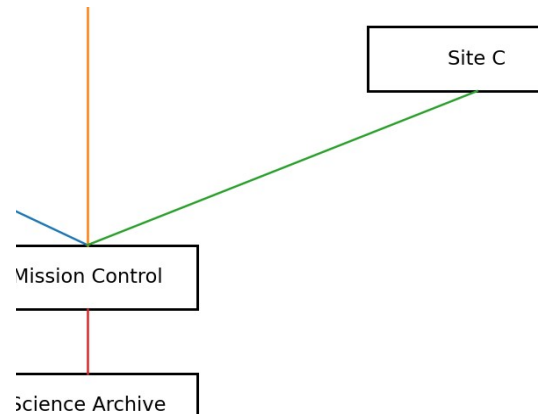


FIGURE 3 – NOTIONAL ADCS AND PROPULSION COMPONENTS

3.1.7 TT&C

The Telemetry, Tracking and Command subsystem shall provide command, telemetry, navigation tracking, time synchronization, and science-data communications over the interplanetary cruise and Qoralis operational phases.

- o QO-1 uses an electrically steered high-gain antenna with a 1.6 m aperture and a redundant low-gain antenna pair. QS-1 uses a fixed 0.55 m patch/reflector assembly for surface-to-orbiter relay and a low-gain antenna for direct emergency communications.
 - Reference deep-space communications parameters
- o Ka-band science downlink peak target: ≥ 2 Mbps at 3.35 AU equivalent range, with adaptive coding and an end-to-end link margin target of ≥ 3 dB.
- o X-band command/telemetry link: 8 kbps minimum command rate and 32 kbps minimum engineering telemetry rate under nominal cruise geometry.
 - QO-1 shall provide at least 4 hours/day of scheduled communications opportunity during nominal target operations. QS-1 shall maintain at least 6 relay contacts per Qoralis rotation, each ≥ 12 minutes, during the nominal surface phase.
 - Ground communications are assumed to use three fictional deep-space antenna sites operated under a common service contract; final antenna count and allocation are to be closed by the link budget.
- o A pair of low-gain antennas on QO-1 provide broad coverage during safe mode. The high-gain antenna is used for science-data transmission and precision radiometric tracking.
- The QISE data link budget shall be reconciled with the science-data generation case of approximately 8 GB/day for QO-1 and 1.2 GB/day for QS-1 before compression.



3.2 PAYLOAD

Source: QISE synthesized mission architecture and fictional instrument trade study

Each payload complement consists of:

- A low-frequency radar sounder
- A thermal-infrared radiometer
- A laser altimeter / optical navigation package
- A surface geophysical instrument suite including a broadband seismometer and heat-flow package

3.2.1 Radar Sounder

The radar sounder provides the primary subsurface-observation capability for QISE. It transmits chirped RF pulses toward Qoralis and measures returned echoes to identify dielectric interfaces, buried fractures, and layer boundaries.

The preliminary radar operates at 8–20 MHz with 100 kHz–2 MHz selectable chirps. Target vertical resolution is ≤ 50 m in low-loss ice and ≤ 150 m in high-loss material; reference orbital swath is ~ 120 km.

Synthetic reference radar instrument

Parameter	Synthetic Radar Reference
Frequency range	8–20 MHz
Chirp bandwidth	100 kHz–2 MHz
Vertical resolution	≤ 50 m nominal; ≤ 150 m high-loss case
Investigation depth	≥ 5 km target
Mass	34 kg
Average power	95 W
Peak transmit power	180 W

Antenna	Dual deployable nadir-facing panels
Data rate	Up to 6 Mbps raw; compressed science stream TBD
Operating mode	Global mapping / targeted fracture transects
Qualification	Flight qualification required

The fictional radar has a preliminary mass of 34 kg, average power of 95 W, and peak transmit power of 180 W. Final antenna geometry and thermal design remain subject to payload trade.

The final radar architecture shall close the 5 km investigation-depth target, ≤ 50 m nominal vertical resolution, pointing, electromagnetic compatibility, mass, power, and data-rate constraints.

3.2.2 The thermal-infrared radiometer measures surface-temperature gradients and localized thermal anomalies associated with internal heat flow.

The radiometer covers a synthetic 7–14 μm band with ≤ 0.15 K NETD at 250 K and ≤ 250 m ground sampling from the reference science orbit.

The fictional radiometer has a preliminary mass of 18 kg, average power of 42 W, and peak power of 60 W; calibration uses an onboard reference and deep-space views.

Parameter	Preliminary QISE Requirement / Target
Spectral range	7–14 μm
Detector technology	Fictional cooled microbolometer array
NETD	≤ 0.15 K at 250 K
Ground sampling	≤ 250 m at reference orbit
Field of view	4.0° cross-track
Frame rate	1–10 Hz selectable
Interface	QO-1 payload-data interface
Mass	18 kg

Power consumption	42 W average / 60 W peak
Data rate	≤1.5 Mbps after onboard selection
Qualification	Flight qualification required

3.2.3 Payload electronics interface the instruments with spacecraft C&DH and provide control, readout, timing, formatting, compression, buffering, health monitoring, and science-packet transfer.

Parameter	Preliminary QISE Requirement / Target
Primary function	Instrument control and science-data handling
Detector interface	Instrument dependent
Spacecraft interface	C&DH payload-data interface
Data processing	Formatting, compression and event selection
Data buffering	Required; ≥32 GB dedicated payload buffer on QO-1
Timing	≤1 ms synchronization for surface geophysics data
Health monitoring	Required
Mass	≤12 kg QO-1 payload electronics; ≤8 kg QS-1
Power consumption	≤35 W QO-1 average; ≤20 W QS-1 average
Data throughput	≥10 Mbps internal payload interface
Radiation tolerance	Suitable for 3.35 AU cruise and Qoralis orbit

3.2.4 Onboard Calibration and Timing

The payload complement includes internal calibration references and mission time transfer to maintain traceability between orbiter and lander datasets.

Radar receiver/timing calibration occurs at least once per 24 h science cycle; radiometer reference observations every 48 h; the lander seismometer time-tag accuracy is ≤1 ms.

Parameter	Preliminary QISE Requirement / Target
Calibration type	Radiometric, geometric, timing and navigation
Radar calibration	Internal receiver/timing calibration every 24 h science cycle

Thermal calibration	Reference observations every 48 h
Seismometer calibration	Instrument self-test every 7 days; ≤ 1 ms time-base verification
Altimeter calibration	Deep-space / target-body reference observations before mapping
Calibration frequency	Mode-dependent; minimum frequencies above
Instrument monitoring	Required
Long-term degradation monitoring	Required
Ground-processing interface	Calibration metadata incorporated into product generation
Mass	Included in instrument allocations

4 LAUNCH SEGMENT DESCRIPTION

The preliminary combined QISE launch-stack mass is 1,460 kg, including 1,050 kg flight hardware, 170 kg propellant, 90 kg launch adapter and separation hardware, and 150 kg allocation for mass growth and late configuration margin. Final launch-vehicle sizing shall use verified flight mass properties plus the launch provider's required performance margin.

The QISE orbiter and lander will be launched as a single combined stack on a fictional commercial launch service and injected onto the reference heliocentric transfer. Vantor Launch Services is the current fictional launch-service organization in the baseline; final launch-vehicle selection remains subject to trajectory, fairing, separation, performance, and schedule analysis.

Launch insurance, if purchased, shall be treated as a separate cost element. The reference estimate shall also separate launch-site processing, range services, spacecraft transportation, and mission-specific separation hardware.

Three-site deep-space communications and data network

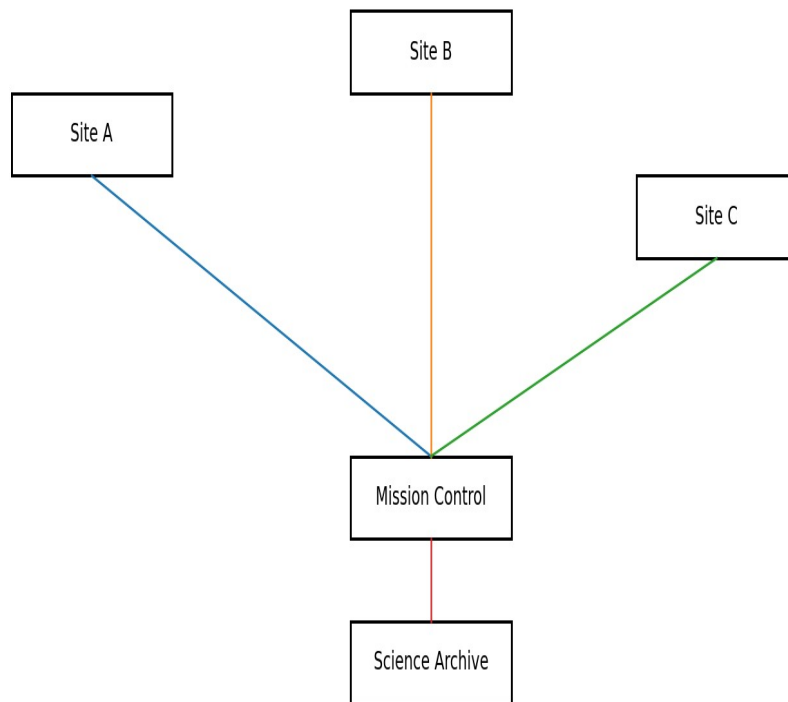


FIGURE 4 – QISE LAUNCH CONFIGURATION

Alternate fictional launch services may be considered if required by availability, injection energy, schedule, cost, or technical compatibility.

- The launch-service down-select shall preserve compatibility with the 1,460 kg combined stack, 4.8 m-class fairing envelope, launch-site environmental constraints, spacecraft separation interfaces, and the required heliocentric injection energy.
- The launch campaign is planned to begin after flight hardware acceptance and shall be completed within 7 months. The baseline launch-readiness date is 1 February 2031.
- Launch preparation shall include transportation, launch-site inspection, final functional checks, battery conditioning, software configuration verification, propellant servicing if applicable, integrated stack testing, and readiness reviews.
- Post-launch operations shall transition into a 30-day early-cruise checkout followed by long-duration cruise operations and scheduled trajectory-correction campaigns.

5 GROUND SEGMENT AND MISSION OPERATIONS

The proposed baseline solution is for NSEO to provide mission authority and flight operations while Quarry Mission Data provides and integrates the deep-space communications interfaces, mission planning tools, navigation support, science-data processing, calibration pipelines, archive, and secure dissemination.

Ground Segment Element	Description
Mission Planning System	Generates time-tagged spacecraft and instrument sequences using ephemeris, power, thermal, communications, navigation and science constraints.
Spacecraft Control System	Command, telemetry, health monitoring, software configuration and autonomous-sequence management for QO-1 and QS-1.
Deep-Space Receiving Stations	Three fictional antenna sites supporting command, telemetry, radiometric tracking and Ka-band science-data reception.
Flight Dynamics	Trajectory determination, orbit propagation, target-body navigation, manoeuvre design and landing-event support.
Data Processing, Archiving and Dissemination	Science packets are reconstructed, calibrated, quality-checked, archived and distributed to authorized NSEO/PGD users.
Data Volume / Availability	Planning case 9.2 GB/day combined; processing ≥ 100 GB/day; scheduled ground availability $\geq 98\%$.

FIGURE 5 – QISE GROUND MISSION NETWORK

At a nominal combined science-data generation rate of 9.2 GB/day, a 30-day communication outage would generate up to 276 GB before compression and prioritization. A preliminary 1 TB mission archive per spacecraft therefore provides more than 3.6 times the 30-day uncompressed outage volume for QO-1 and QS-1 combined.

The current planning case is 8 GB/day from QO-1 and 1.2 GB/day from QS-1. Ground processing is sized at ≥ 100 GB/day of raw-equivalent science-data processing capacity to support reprocessing, calibration, navigation updates, and product generation. Priority science products shall be available within 72 hours of receipt.

6 SYSTEMS ENGINEERING AND PROJECT MANAGEMENT APPROACH

6.1 QISE WILL USE HERITAGE EVIDENCE WHERE APPLICABLE, BUT THE FINALIZED BASELINE IS NOT ASSUMED TO INHERIT THE TRL OF ANY REFERENCE SPACECRAFT OR INSTRUMENT. MAJOR SPACECRAFT-BUS COMPONENTS ARE TARGETED TO REACH TRL ≥ 8 BEFORE FLIGHT QUALIFICATION. CRITICAL PAYLOAD TECHNOLOGIES SHOULD BE AT LEAST TRL 5 AT THE START OF B2 DEVELOPMENT AND TRL ≥ 7 BEFORE THE START OF SPACECRAFT-LEVEL QUALIFICATION. ANY REUSED TEST EVIDENCE MUST BE DEMONSTRATED APPLICABLE TO THE QISE CONFIGURATION, ENVIRONMENT, INTERFACES, AND OPERATING ASSUMPTIONS.

6.2 PHASING LOGIC AND NAMING CONVENTIONS

For the QISE cost baseline, the PM nomenclature is B1, B2, C, D, E, and F. The nominal schedule is: B1 = 13 months (September 2026–October 2027); B2 = 9 months (October 2027–July 2028); C = 12 months (July 2028–July 2029); D = 19 months (July 2029–February 2031, including manufacturing, integration, test, launch and commissioning); E = 30 months (April 2031–October 2033, including the 4.2-year cruise that began after launch is treated within D/E transition assumptions); F = approximately 2 months for disposal/close-out. For cost modelling, the total program window from B1 start through nominal science completion is 85 months, followed by a separate approximately 2-month Phase F close-out period.

PREP 1	Prototyping Activities (before Gate 1)
PREP 2	Prototyping Activities (during Phase 0/A)
Phase 0	Concept & Feasibility Studies
Phase A	System Definition
Phase B	Preliminary Design
Phase C	Detailed Design
MAIT	Manufacturing, Assembly, Integration and Test (of Systems, Subsystems and Components)
SAIT	Segment Level Assembly, Integration and Test
LEOP	Launch and Early Operations (including deployment and commissioning)
Phase E	Cruise and Nominal Science Operations
Phase X	Extended Science Operations

Phase F	Decommissioning and disposal
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6.3 NSEO PROJECT MILESTONES

The finalized PM schedule is the nominal baseline for this CADRe Part A. It establishes payload, orbiter, lander, and launch milestones used to determine NSEO need dates and is consistent with the current QISE phase-duration and PM scenario baseline.

Instrument / PM Phase	Gate / Milestone	Major Milestone	Milestone Date
B1	MCR	Mission Concept Review	29 March 2027
B1	CA	Contract Award	1 October 2027
B2	PAY-PDR	Payload Preliminary Design Review	1 April 2028
B2	BUS-PDR	Bus Preliminary Design Review	1 July 2028
B2	PDR	Overall Preliminary Design Review	1 July 2028
C	PAY-CDR	Payload Critical Design Review	1 January 2029
C	BUS-CDR	Bus Critical Design Review	1 April 2029
C	CDR	Overall Critical Design Review	1 April 2029
D	MRR	Manufacturing Readiness Review	1 June 2029
D	LT-READY	Long-Lead Hardware Ready	1 October 2029
D	FLATSAT	Integrated Avionics Flatsat Test	1 February 2030
D	FH	Flight Hardware Complete	1 September 2030
D	ETRR	Environmental Test Readiness Review	1 October 2030
D	AR/SAR	Acceptance Review	1 December 2030
D	LRR	Launch Readiness Review / Flight Stack Complete	1 January 2031
D	L	Launch	1 February 2031
D	ECO	Early Cruise Checkout Complete	1 March 2031

Instrument / PM Phase	Gate / Milestone	Major Milestone	Milestone Date
D	ARR	Arrival Readiness Review	1 November 2034
D	OI	QO-1 Orbit Insertion	1 December 2034
D	LAND	QS-1 Surface Landing	19 December 2034
E	SCI-START	Nominal Science Operations Start	1 January 2035
E	EOM	End of Nominal Science Operations	1 January 2037
F	EOL	Phase F disposal / close-out end	1 March 2037

6.4 SAFETY AND MISSION ASSURANCE

Initial assessment: Mission Class B. The program requires formal mission assurance, qualification, acceptance, verification, configuration control, navigation validation, landing readiness reviews, and independent science-data quality checks. Safety and mission-assurance effort is a direct cost and schedule driver because QO-1 and QS-1 have different environmental and operational verification profiles.

6.5 QUALIFICATION AND ACCEPTANCE STRATEGY, MODEL PHILOSOPHY

The QISE qualification and acceptance strategy will verify that QO-1 and QS-1 are capable of surviving launch and performing their required functions throughout cruise, target operations, and the planned surface mission. The approach uses one Engineering Qualification Unit for common avionics and communications interfaces, followed by two flight vehicles with independent acceptance verification.

The qualification unit will represent the flight avionics and communications configuration as closely as practical and will undergo functional, vibration, thermal-vacuum, electromagnetic compatibility, radiation-screening, and end-to-end command/data testing. The landing subsystem will also use a dedicated structural/landing test article for descent and touchdown verification. Existing qualification evidence may be reused only where the QISE configuration and environment are demonstrated equivalent.

Following successful qualification, QO-1 and QS-1 will be manufactured using the controlled design baseline. Each flight vehicle will undergo its own functional, electrical, communications, software, payload, and applicable environmental acceptance testing. QS-1 will additionally complete a representative landing-sequence rehearsal and surface deployment test.

The qualification and acceptance program also accounts for the logistics of a multi-year deep-space mission. The two flight vehicles will be integrated as a launch stack, transported under

controlled environmental conditions, tested at the launch site, and maintained in a controlled configuration through final readiness activities. Cruise support will include scheduled trajectory-correction windows and independent command-load validation.

Following launch, the combined stack will undergo a 30-day early-cruise checkout. Target-arrival operations begin approximately 45 days before Qoralis encounter. QO-1 orbit insertion and QS-1 landing will be separated by approximately 18 days to allow navigation and health assessments between critical events.

All qualification and acceptance results will be documented through the QISE verification and configuration-management process. Any anomaly identified during testing, transportation, launch-site processing, cruise, arrival, or commissioning will be assessed and resolved or dispositioned before the affected mission phase proceeds. Final approval for launch and science operations will be based on completion of required verification and successful Launch Readiness, Arrival Readiness, and Operations Readiness Reviews.

6.6 THE CURRENT QISE PROGRAM RISK REGISTER AND RISK BREAKDOWN STRUCTURE ARE THE CONTROLLING PROGRAMMATIC RISK REFERENCES. COST AND SCHEDULE RISKS OF PARTICULAR IMPORTANCE TO PART A ARE LONG-LEAD AVIONICS AND RADAR DEVELOPMENT, PAYLOAD-TO-BUS ELECTROMAGNETIC COMPATIBILITY, DEEP-SPACE COMMUNICATIONS PERFORMANCE, TRAJECTORY CORRECTION REQUIREMENTS, PRECISION LANDING, TARGET-BODY NAVIGATION UNCERTAINTY, THERMAL-CONTROL PERFORMANCE, RADIATION EXPOSURE, GROUND-NETWORK AVAILABILITY, AND EXTENDED CRUISE STAFFING.

The project team shall produce a risk register (ref CADRe Part C).

6.7 ORGANIZATIONAL BREAKDOWN STRUCTURE

NSEO Project Team: mission authority and program manager; responsible for systems engineering, payload authority, program management, mission integration, mission planning, mission assurance, and government flight operations.

- PGD: principal science authority and operational science user; provides measurement priorities, geophysical models, calibration requirements, and science-product acceptance criteria.
- Payload: Blue Meridian Instruments; payload electronics and instrument integration: Heliox Orbital Systems / Cinderline Surface Systems as applicable.
- Radar and thermal instruments: Blue Meridian Instruments.
- Orbiter bus: Heliox Orbital Systems, selected fictional spacecraft contractor.
- Ground segment: Quarry Mission Data, responsible for provision and integration of the deep-space communications interfaces, mission-planning tools, navigation support, processing, and archive.

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- Launch: Vantor Launch Services. Surface lander: Cinderline Surface Systems. Science Segment: fictional university laboratories contracted through PGD. No international mission partner is assumed in the baseline.

APPENDICES

A ACRONYM LIST

Acronym	Definition
CADRe	Cost Analysis Data Requirement
NSEO	National Space Exploration Office
PGD	Planetary Geophysics Directorate
QISE	Qoralis Interior and Surface Explorer
QO-1	Qoralis Orbiter 1
QS-1	Qoralis Surface Geophysics Lander 1
C&DH	Command and Data Handling
ADCS	Attitude Determination and Control System
TT&C	Telemetry, Tracking and Command
MPS	Mission Planning System
SCS	Spacecraft Control System
MCR	Mission Concept Review
PDR	Preliminary Design Review
CDR	Critical Design Review
MRR	Manufacturing Readiness Review
ETRR	Environmental Test Readiness Review
AR/SAR	Acceptance Review / System Acceptance Review
LRR	Launch Readiness Review
ARR	Arrival Readiness Review
LEOP	Launch and Early Operations Phase
MAIT	Manufacturing, Assembly, Integration and Test
SAIT	Spacecraft/Segment Assembly, Integration and Test
TRL	Technology Readiness Level
EQU	Engineering Qualification Unit
FM	Flight Model
NETD	Noise-Equivalent Temperature Difference
GSD	Ground Sample Distance
ΔV	Change in velocity
